

Attention from Action, for Action: Emergent Visual Bottlenecks for Policy Learning

Zheyu Zhuang¹ ✉, Ruiyu Wang¹, Nick Heppert², Johannes Fabian Hahn³,
Abhinav Valada², Florian T. Pokorny¹, Danica Kragic¹

¹Department of Robotics, Perception and Learning, KTH Royal Institute of Technology, Sweden

²Department of Computer Science, University of Freiburg, Germany

³Department of Informatics, Universität Hamburg, Germany

Abstract: Visual bottlenecks that focus policy inputs on regions of interest (ROIs) can improve data-efficient visuomotor learning by separating *where to look* from *how to act*. Many ROI interfaces rely on external spatial labels, such as gaze, object classes, or affordance annotations. Label-free alternatives often derive crops from trajectories by detecting gripper or motion events and centering a fixed crop at the projected end-effector. Such action-derived crops are useful spatial priors that require no additional labels, but they encode fixed choices about event timing, proxy points, and crop scale. When the visual evidence needed for control lies away from the end-effector or changes continuously with task progress, these crops can become misaligned. We propose Seeker, a task- and state-conditioned readout that learns *attention from action*. Starting from frozen DINO features, Seeker iteratively updates a query with gathered visual evidence, producing progression-aware ROIs solely from action supervision. The learned ROI serves as a spatial interface for RGB cropping, mask-guided background augmentation, and point-cloud filtering. In simulation and the real world, Seeker improves data efficiency and robustness over no-crop, augmentation, and action-derived crop baselines. On real robots, Seeker raises average in-domain success from the best baseline’s 48.3 to 76.7% and success under lighting/background shifts from 20.0 to 60.0%.  [CODE](#)

Keywords: Imitation Learning, Data Efficiency, Region of Interest (ROI)

1 Introduction

Imitation learning is increasingly applied to long-horizon manipulation in visually complex environments, where decision-critical cues are often sparse and localized [1]. A common way to improve data efficiency is to separate *where to look* from *how to act*: estimate a region of interest (ROI) and train the policy on the resulting focused observation [2, 3, 4]. The difficulty is that the ROI is a control-dependent visual bottleneck rather than a fixed semantic category. It can shift with robot state, task phase, and camera view, and may correspond to an object, an affordance region, a tool–object contact, a goal region, or a spatial relation between objects.

Because this control-dependent ROI is hard to specify directly, existing interfaces often rely on external spatial cues or semantic priors, such as predicted human gaze [2], object- or tool-centered crops [4], or vision–language model (VLM) grounding [5, 6, 7]. These signals can be powerful, but are not typically available in imitation datasets and may not directly specify the visual evidence needed for the next control decision. For long-horizon tasks, providing such evidence through language or VLMs often requires detailed stage-aware prompts or subtask decomposition [8, 9].

In contrast, the action stream is intrinsic to imitation learning and tied to visual relevance: actions are generated from the cues the demonstrator used to decide what to do next. This makes the observation–action stream an implicit but control-aligned source of supervision for where to look. Action-grounded heuristics exploit this connection by deriving spatial targets from demonstration

trajectories, for example by projecting the end-effector at keyframes detected from motion stops or gripper-state changes [3, 10, 11]. These methods bring localization closer to the action signal, but still hand-design the action-to-ROI mapping through event thresholds, proxy points, and crop scales. This fixed design can become limiting under trajectory and task variability: velocity or acceleration changes can create noisy keyframes, continuous interactions may lack reliable discrete events, and the relevant cue may lie away from the proxy, such as on a tool contact, object part, goal region, or spatial relation.

We propose Seeker to learn this action-to-ROI mapping directly from observation–action streams. Like action-grounded heuristics, it requires no spatial labels; unlike them, it does not predefine trajectory events, proxy points, or crop scales. Seeker starts from a task- and state-conditioned query over frozen DINOv3 [12] patch features and iteratively refines this query through visual evidence gathered from image patches. This readout is therefore not a static crop predictor: its focus can shift as the task stage and robot state change. Trained through action prediction, the resulting ROI adapts in size, location, and focus as control demands evolve, moving between object-, contact-, and relation-centric evidence rather than following semantic labels or keyframe heuristics.

We evaluate Seeker in simulation and real-world manipulation to test three claims about action-supervised visual bottlenecks:

- (1) **Action supervision can recover policy-useful ROIs:** without spatial labels, Seeker nearly matches the privileged Oracle ROI reference for policy learning.
- (2) **The resulting bottleneck improves policy learning:** under the same downstream RGB stack, Seeker raises average simulation success from 42.6% to 62.6%, and improves real-world in-domain success from 48.3% to 76.7% over the best baseline.
- (3) **The learned ROI is reusable and supports robustness:** frozen Seeker ROIs support mask-guided augmentation and point-cloud filtering, and improve real-world robustness under both lighting and background shifts, raising average shifted-condition success from 20.0% to 60.0%.

2 Related Work

Spatial Bottlenecks from Priors and Auxiliary Signals. Spatial bottlenecks restrict policy perception to task-relevant regions, reducing the visual search space faced by downstream control. Prior work obtains such regions from external annotations or auxiliary structure, including human gaze [2], object- or tool-centered crops [4], VLM grounding [5, 6], recurring keypoints in demonstrations [13], or keypoints learned jointly with the policy [14]. However, a bottleneck tied to an external cue, semantic grounding, or keypoint representation may capture a plausible task region without matching the control-dependent evidence needed for the next action.

Action-Derived Spatial Bottlenecks. A closely related line converts demonstrated actions into spatial targets. Q-attention [15] and Coarse-to-Fine Q-attention [3] discretize action localization and zoom into relevant regions, while PerAct [10], RVT [16], and RVT-2 [11] use keyframe gripper/TCP targets in voxel or multi-view representations, with RVT-2 explicitly predicting an area of interest before zoomed-in pose estimation. CLAP [8] adds VLM grounding and subtask decomposition. Although these methods differ architecturally, their spatial bottlenecks are still shaped by engineered choices about when to localize, which action proxy to use, and what crop, zoom, or voxel scale to impose. Seeker instead learns this action-to-ROI readout from observation–action streams, producing an explicit ROI reusable for RGB cropping, mask-guided augmentation, and point-cloud filtering.

Data-Efficient Policy Learning. Spatial bottlenecks are complementary to policy-level strategies for data efficiency. Reflection-consistent augmentation [17] expands the effective training distribution through symmetry-based image–action transforms, while equivariant policy learning [18, 19] builds geometric structure into the policy architecture. These methods improve the policy learner or training distribution, but do not explicitly decide which visual evidence should be preserved for control. By contrast, an action-grounded bottleneck changes the visual input itself, making it compatible with standard policies while remaining complementary to augmentation and architectural priors.

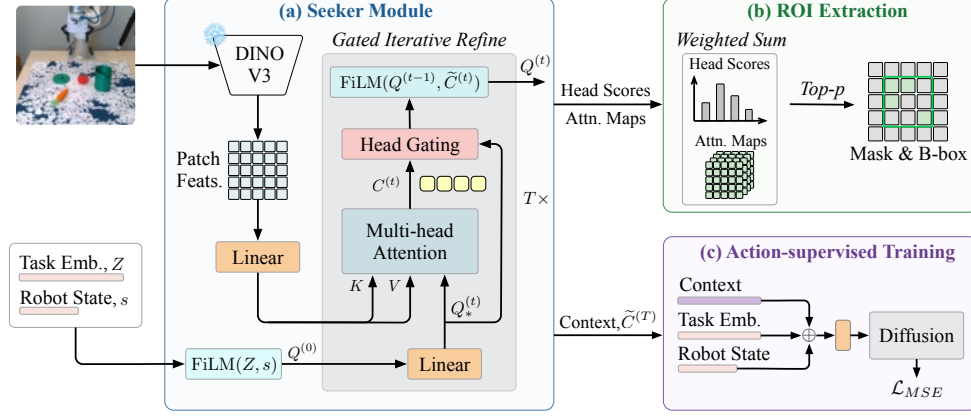


Figure 1: **Seeker Overview.** (a) A state-conditioned query is iteratively refined over frozen DINOv3 patch tokens, producing a compact visual context together with attention maps and head scores. (b) Head-score-weighted attention is converted into a mask and bounding box using a `top-p` mass criterion. (c) The visual context is trained through a diffusion action-prediction loss, allowing the ROI to emerge without spatial supervision.

3 Methodology

Fig. 1(a) illustrates Seeker, a lightweight readout over frozen DINOv3 [12] patch tokens. Its design is related to attention-probing readouts over pretrained visual features [20, 21, 22], but is adapted for action-grounded ROI discovery. Rather than performing a single visual lookup, Seeker treats readout as an iterative search process: a task- and state-conditioned query repeatedly attends to DINO patches, gathers visual context, and updates itself before the next lookup. This allows the ROI to move from an initial control-conditioned prior toward visual evidence relevant to the current action. To capture diverse spatial cues, Seeker splits the projected DINO patch features into multiple readout heads and uses query-dependent head gating, allowing different heads to dominate as the task stage and robot state change. The final readout produces both a compact visual context and an ROI attention map; details and visualizations are provided in App. A.1.

3.1 Seeker Architecture

Robot-conditioned Query. Given frozen DINOv3 [12] patch features $x \in \mathbb{R}^{N_p \times D}$, Seeker first projects them into attention keys and values $K, V \in \mathbb{R}^{N_p \times D}$. The query is task- and state-conditioned, and represents *what to look for now*. Specifically, given a task embedding $Z \in \mathbb{R}^D$ and proprioceptive state $s = [s_{\text{eef}}, s_{\text{grip}}] \in \mathbb{R}^S$, the initial query is

$$Q^{(0)} = \text{FiLM}(Z, s) = (1 + \gamma(s)) \odot Z + \beta(s),$$

where $\gamma(\cdot)$ and $\beta(\cdot)$ are predicted by an MLP, and $\odot$ denotes element-wise multiplication.

Iterative Gated Readout. Seeker applies the same gated cross-attention readout for T refinement steps. At each step, the current query is first projected and then attends to the visual keys and values:

$$Q_* = \phi_q(Q), \quad \{A_h, C_h\}_{h=1}^H = \text{MHA}(Q_*, K, V),$$

where $\phi_q(\cdot)$ is a learnable linear projection, and A_h and C_h are the attention map and context from head h . Instead of uniformly averaging heads, Seeker predicts query-dependent head weights. Let $\bar{Q}_*$ denote the mean of the head-wise query components. With dot product $\langle \cdot, \cdot \rangle$ and learned projection $\phi_g(\cdot)$, each head is scored as

$$\omega_h = \text{softmax}_h(\langle C_h, \phi_g(\bar{Q}_*) \rangle).$$

The head weights fuse the per-head outputs and update the query:

$$\tilde{C} = \sum \omega_h C_h, \quad \tilde{A} = \sum \omega_h A_h, \quad Q \leftarrow \text{FiLM}(Q, \tilde{C}).$$

After T applications, the final $\tilde{C}$ is used as Seeker’s compact visual context, and the final $\tilde{A}$ is reshaped to the DINO patch grid as the ROI map. Head-gating visualizations in App. A.1 show that head weights vary with robot state and task stage.

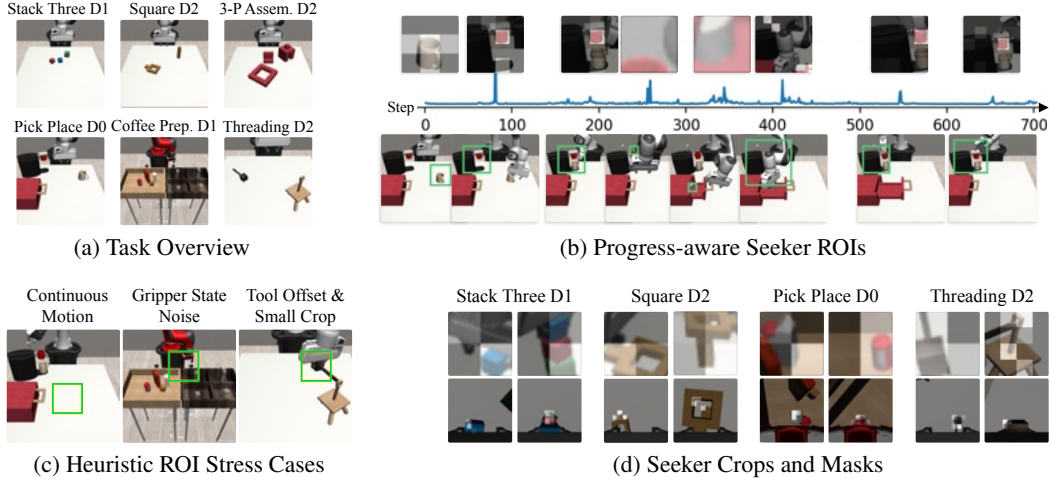


Figure 2: **Simulation setup and Seeker visualization.** (a) MimicGen [24] task overview. (b) In Coffee Preparation, Seeker ROIs shift with task progress; selected frames correspond to peaks in pairwise context-feature distance. (c) RVT2-style heuristic ROI stress cases: missed continuous-motion cues, noisy gripper-triggered keyframes, and tool-contact regions offset from the TCP proxy. (d) Seeker crops and masks across tasks.

Action-supervised ROI Interface. Seeker turns action supervision into an explicit ROI interface without spatial labels. As shown in Fig. 1(c), the final context $\tilde{C}$ is concatenated with proprioception and a task embedding, then passed to a diffusion action head trained with the standard noise-prediction objective [23]. This head is used only to provide a dense action-supervised signal for ROI discovery, not as the final downstream controller. After multi-task Seeker training, we discard the diffusion head and freeze the Seeker readout for downstream policies.

As shown in Fig. 1(b), the final gated attention map $\tilde{A}$ is converted into a patch-level ROI by selecting the smallest set of tokens whose cumulative mass exceeds a top_p threshold. This produces a tight bounding box and a coarse patch mask, which are reused in three downstream forms. **RGB cropping:** the box crops and resizes the static third-person view, providing a local visual input for RGB policies. Because cropping removes absolute spatial information, we FiLM-condition policy features on the box location and scale. We do not crop eye-in-hand views, whose scale and content change rapidly near contact. **Point-cloud filtering:** the same image-plane box filters projected point-cloud points before downsampling, yielding a localized 3D input without retraining Seeker. **Mask-guided augmentation:** the patch mask preserves task-relevant regions while perturbing task-irrelevant appearance, improving robustness to background changes. All ROIs are precomputed offline, so downstream policy training incurs no additional Seeker cost. Training protocols and implementation details are provided in App. A.2.

4 Simulation Evaluations

Experimental Protocol Fig. 2a illustrates the six MimicGen[24] tasks used in our simulation study: Stack Three D1, Square D2, Three-Piece Assembly D2, Coffee Preparation D1, Pick & Place D0, and Threading D2. These tasks cover high spatial variance and long-horizon manipulation; detailed task descriptions are provided in App. B. We omit D0/D1/D2 suffixes where clear.

We train a **single** multi-task Seeker once, using 100 demonstrations per task from the original-background RGB data, and keep it frozen for all simulation experiments. This design isolates the ROI interface from downstream policy learning: Seeker remains unchanged, while each policy is trained with the input format and data protocol required by its evaluation setting. We evaluate this fixed ROI interface through RGB data efficiency, background generalization, and point-cloud transfer, testing whether action supervision yields a reusable spatial bottleneck across input protocols, appearance shifts, and sensing modalities.

	Stack Three	Square	3-P Assembly	Coffee Prep.	Pick & Place [†]	Threading	Avg.
<i>Input-level Comparison</i>							
DiffPo (Pre)	52.7	22.0	14.0	65.3	20.0	20.7	32.5
MirrorAug	68.0	32.7	21.0	62.0	20.7	18.7	37.2
RVT2-Crop	71.3	26.0	26.7	69.3	49.3	12.7	42.6
Seeker (ours)	81.3	46.0	58.7	82.0	69.3	38.0	62.6
<i>Gain (rel., %)</i>	<i>14.0_↑</i>	<i>40.7_↑</i>	<i>119.9_↑</i>	<i>18.3_↑</i>	<i>40.6_↑</i>	<i>83.6_↑</i>	<i>52.8_↑</i>
<i>External SOTA and Privileged ROI References</i>							
RAVEN	80.7	50.0	26.7	72.7	54.7	28.0	52.1
Oracle ROI*	80.7	44.7	62.7	81.3	71.3	44.7	64.2

Table 1: **Data Efficiency on MimicGen with 100 Demos.** Success rate (%) averaged over three seeds; stds in App. D.1. Gray cells mark the strongest non-Seeker input-level baseline. Seeker improves average success by 20 points over this baseline (62.6 vs. 42.6), with a mean task-wise relative gain of +52.8%. It also exceeds external full-policy SOTA RAVEN (62.6 vs. 52.1) and remains within 1.6 points of the privileged Oracle ROI. *The oracle provides stage-aware affordance ROIs. [†]Pick & Place uses strict episode success with all baselines rerun.

Baseline Design. (1) *RGB Policy Learning.* The RGB comparison isolates Seeker’s action-grounded visual bottleneck under a shared downstream stack. All controlled baselines use the same pretrained ResNet-18 policy encoder, since pretrained features substantially improve RGB manipulation performance [19]. In addition to vanilla Diffusion Policy [23], MirrorAug [17] provides an augmented no-crop reference, applying reflection-consistent image–action augmentation without an explicit spatial bottleneck. RVT2-Crop is our RVT-2-inspired [11] crop baseline, adapting its heatmap-based localization interface to predict a 2D crop target from frozen DINOv3 features and proprioception. The heatmap is supervised using the original keypoint extraction recipe based on gripper-state changes and low end-effector velocity, testing whether this strong action-derived prior is sufficient for policy cropping. Oracle ROI uses privileged stage-aware affordance pixels to crop target objects; it is not a control upper bound because contacts or inter-object relations may lie outside the crop. RAVEN [19] is included as an external equivariant-policy SOTA reference outside the controlled stack.

(2) *Background Generalization.* This setting tests whether the same Seeker trained on original-background demos remains useful under shuffled tabletop textures (Fig. 3b) by using its mask to guide appearance augmentation. We use Random Overlay[17] as the default augmentation for visual domain generalization, blending each observation with a texture image using a spatially constant preservation mask. Guided variants replace this constant mask with a spatial mask when available: RVT2-Crop uses its Gaussian heatmap for the front view, while Seeker-Guided uses frozen Seeker masks for both views.

(3) *Point-Cloud Filtering.* This setting tests whether Seeker’s ROI functions as a spatial interface rather than an RGB-specific feature. The baseline is DP3[25] with manual workspace cropping, following common practice in point-cloud policy learning [26, 25, 18]. Seeker is not retrained for point-clouds. Its frozen RGB ROI filters 3D points by image-plane projection before the same DP3.

RGB Data Efficiency. We evaluate Seeker as an input-level visual bottleneck under the controlled RGB policy stack. Tab. 1 shows that Seeker improves average success from 42.6% for the strongest non-Seeker input-level baseline to 62.6%, with a mean task-wise relative gain of +52.8%. RVT2-Crop is the strongest non-Seeker baseline on average, but its uneven gains reflect settings where fixed keyframe/proxy assumptions can become limiting: as shown in Fig. 2c, they can miss continuous-motion cues, produce noisy gripper-triggered targets, or focus on TCP proxies offset from the interaction region. Seeker improves consistently, with the largest gains on spatially ambiguous or contact-rich tasks such as Three-Piece Assembly, Pick & Place, and Threading, where policies must repeatedly localize small control-relevant regions across objects, tools, and task stages. This supports the central premise that action supervision can expose visual evidence useful for control, rather than only producing semantic or object-centric crops. Seeker also exceeds the external SOTA reference RAVEN on average (62.6 vs. 52.1) and nearly matches Oracle ROI (64.2), despite using no scripted affordance boxes, spatial labels, gaze, or language grounding.

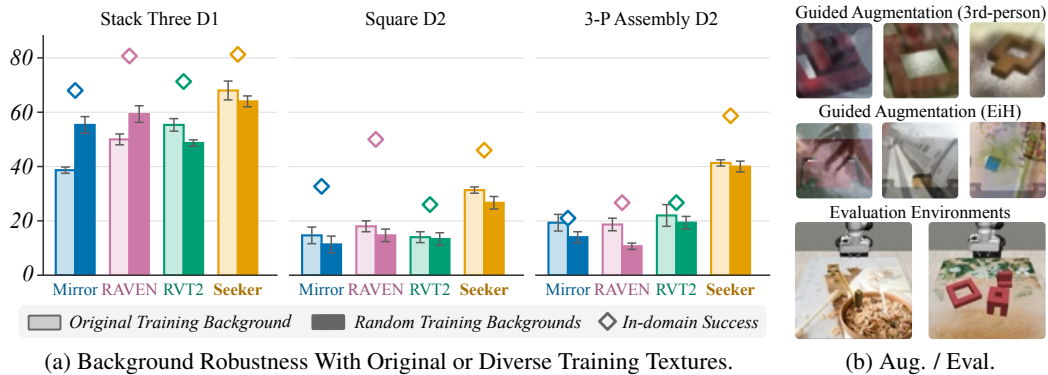


Figure 3: **Background Robustness and Guided Augmentation.** (a) Light bars use original-background training, solid bars use 25 randomized-background training, and diamonds show in-domain mean success from the main benchmark. We compare MirrorAug, RAVEN+Overlay, RVT2-Crop, and Seeker-Guided augmentation. (b) Seeker-guided augmentation examples and evaluation environments.



Figure 4: **Point-cloud transfer with DP3 [25].** Left: manual workspace cropping vs. manual+Seeker ROI filtering, with point clouds independently scaled for visualization. Right: success rates with 200 demonstrations for no crop, manual crop, and manual+Seeker filtering; absolute gains are computed over the manual cropping.

Mask-Guided Augmentation for Background Generalization. Beyond data efficiency, we reuse the same frozen Seeker trained on the original training set under background shifts. Its mask controls image superimposition by preserving predicted control-critical regions while perturbing task-irrelevant appearance. Given a Seeker mask $M \in [0, 1]^{H \times W}$, image I , and texture image I_{tex} , we form $I_{\text{aug}} = M \odot I + (1 - M) \odot I_{\text{tex}}$, where M is broadcast over color channels and $\odot$ denotes element-wise multiplication. Random Overlay [17] is the constant-mask case; Seeker-Guided uses the frozen control-aware mask.

We evaluate Stack Three D1, Square D2, and Three-Piece Assembly D2 under shuffled tabletop textures. Policies are trained either on the original 100 demonstrations or on the same trajectories re-rendered with 25 shuffled tabletop backgrounds. Baseline-specific augmentation details are described in Sec. 4; examples and implementation details are shown in Fig. 3b and App. D.3.

Fig. 3a shows that Seeker-Guided augmentation is strongest when using the Seeker trained on the original demonstrations, and remains best on average when combined with background-randomized training. This indicates that the action-grounded regions identified by Seeker transfer across background changes well enough to guide augmentation. Rather than perturbing the whole image, Seeker-Guided augmentation preserves the predicted control-relevant regions and applies appearance variation mainly to the surrounding background, reducing the burden on the policy to learn under visually distracting textures. The mixed effect of background-randomized training across tasks and methods further suggests that visual diversity does not by itself guarantee background generalization, especially when additional texture complexity can obscure fine interaction cues.

Point-cloud Transfer. We evaluate whether Seeker’s image-plane ROI can serve as a plug-and-play spatial filter for point-cloud policies. Given a predicted Seeker box, we remove points whose image-plane projections fall outside the ROI, then apply farthest-point sampling to match the fixed input size required by DP3 [25]. Since point-cloud policy learning commonly relies on manually cropping a task-specific workspace to remove background and tabletop regions [26, 25, 18], we evaluate

Seeker as a complementary ROI filter on top of this manual workspace crop. This transfer study uses 200 demonstrations because, with 100 demonstrations, the manual-crop baseline was too weak to support informative relative comparisons. As shown in Fig. 4, adding Seeker ROI filtering makes the retained point-clouds more task-focused and improves DP3 over manual workspace cropping by 30.3 absolute points on average on the three tasks, despite Seeker being trained only from RGB action supervision. Overall, DP3 remains below image-based policies (Tab. 1), consistent with prior work [18]. Seeker-only filtering and full per-task analyses are provided in App. D.2.

5 Real-world Evaluations

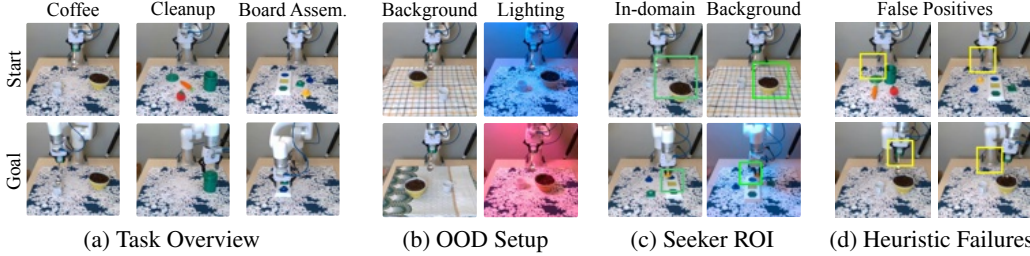


Figure 5: **Real-world Setup and Visual Diagnostics.** (a) Start and goal configurations for the three real-world tasks. (b) OOD evaluation conditions with background and lighting shifts. (c) Seeker ROIs remain focused on predicted control-relevant regions under appearance changes. (d) Heuristic crops can produce ghost keyframes or miss interaction-relevant continuous-motion cues without reliable gripper-state anchors.

Method	Coffee Trans.			Table Cleanup			Board Assem.			Average		
	ID	Light.	Bg.	ID	Light.	Bg.	ID	Light.	Bg.	ID	OOD	Retention
MirrorAug	45	30	0	40	10	10	20	0	20	35.0	11.7	33.4
RVT2-Crop	40	15	0	60	25	35	45	20	25	48.3	20.0	41.4
Seeker (ours)	85	80	35	80	75	60	65	60	50	76.7	60.0	78.2

Table 2: **Real-world experiments.** Success rate (%) over 20 rollouts under in-domain (*ID*), lighting-shift (*Light.*), and background-shift (*Bg.*) evaluations. Gray cells mark the strongest non-Seeker baseline in each column. *Avg. OOD* averages lighting and background shifts; *Ret.* is the retention ratio, *Avg. OOD/Avg. ID*.

We evaluate Seeker and two representative baselines on a real-world multi-view RGB setup with a UFactory xArm7, using a fixed third-person camera and an eye-in-hand camera. The three tasks are performed on visually distracting table textures and require different forms of spatial attention (Fig.5a): *Coffee Transport* is a low-tolerance continuous manipulation task, where the robot must carry coffee beans on a spoon and deposit them into a small white espresso cup under low foreground–background contrast; *Table Cleanup* is a long-horizon sequential insertion task requiring 3D object rotations, including tilting a carrot so that it fits through the bin opening; and *Board Assembly* requires precise shape-based placement despite color mismatch between the objects and target slots. Detailed real-world task descriptions are provided in App. E.

Tab. 2 shows that Seeker improves average in-domain success from the best baseline’s 48.3 to 76.7% and raises average OOD success from 20.0 to 60.0%. Its 78.2% retention indicates that the learned ROI remains usable under real appearance changes, including lighting shifts that over-expose the eye-in-hand view.

The RVT2-Crop results highlight two settings where action-heuristic crop supervision is challenged. First, in Table Cleanup and Board Assembly, inconsistent velocity profiles across demonstrations can create false-positive low-speed keyframes, causing the crop predictor to attend to spurious targets before reaching the relevant interaction region. Second, in Coffee Transport, continuous spoon motion provides no reliable gripper-state anchor, so the heuristic crop can miss the cup–spoon interaction and lock onto rotation-only or final end-effector poses (Fig. 5d). When this crop is misaligned, the policy loses useful third-person context and must rely more on the eye-in-hand stream, which is fragile when the white cup leaves the wrist-camera field of view or is washed out by lighting.

Variant	Stack Three	Square	3-P Assem.	Coffee Prep.	Pick & Place [†]	Threading	Avg.
Low-res Crop	82.7 _{1.4} ↑	46.7 _{0.7} ↑	56.7 _{2.0} ↓	75.3 _{6.7} ↓	70.0 _{0.7} ↑	36.0 _{2.0} ↓	61.2 _{1.4} ↓
FiLM Only	64.7 _{16.6} ↓	29.3 _{16.7} ↓	16.7 _{42.0} ↓	69.3 _{12.7} ↓	28.0 _{41.3} ↓	28.7 _{9.3} ↓	39.4 _{23.2} ↓

Table 3: **ROI Conditioning, Cropping, and Resolution Ablation (100 demos).** Entries report success rate (%); subscripts show absolute percentage-point change w.r.t. Seeker in Tab. 1.

6 Ablations

Training Objective for Seeker. We compare direct regression, flow matching [27, 28], and IMLE [29] for training Seeker from action supervision. Direct regression yields noisy attention, while generative objectives produce more stable ROIs: IMLE is cleaner but diffuse, and flow matching gives tighter masks closest to diffusion. Across 300 trajectories from Square, Stack Three, and Three-Piece Assembly, the average ROI box IoU to the diffusion-trained Seeker is 0.50 for IMLE and 0.67 for flow matching. Additional qualitative comparisons in App. A.2.

ROI Conditioning, Cropping, and Resolution. We next ablate whether Seeker’s gains come from box-level conditioning, explicit spatial cropping, or higher-resolution ROI content. FiLM-only conditions full-image CNN features on the predicted box location and scale, while low-res crop applies the same normalized box to the downsampled policy-resolution image. Tab. 3 shows that low-res cropping nearly matches Seeker on average, whereas FiLM-only is substantially weaker. This suggests that the main benefit comes from exposing the progression-aware ROI as an explicit crop, not merely telling the network where the box is. The small remaining gap indicates that high-resolution ROI content can help in fine-grained stages, but is secondary to reducing full-scene visual search.

7 Limitations

Demonstration Coverage. Seeker can be pretrained on fewer demonstrations than the downstream policy when they contain sufficient spatial variation in control-relevant interaction regions and task phases. In App. C, spatially diverse subsets recover similar ROI behavior and policy performance to the full Seeker pretraining set, suggesting that coverage matters more than raw count. In spatially invariant or visually redundant phases, actions may be predictable from proprioception or trajectory regularities, giving little incentive to localize visual evidence. This shows little in-domain impact in Tab. 1, where Seeker still matches or approaches Oracle ROI on Coffee Preparation and Pick & Place despite largely static targets. New layouts that vary these targets may require additional spatially varied demonstrations or retraining.

Training and Inference Overhead. Seeker adds a separate pretraining stage before the downstream policy, after which it is frozen and reused. For consistency, our main experiments train Seeker on the full policy-training set, giving a conservative total training overhead of roughly 1.5–1.8× over training the downstream policy alone. This overhead can be reduced by training Seeker on fewer spatially diverse demonstrations, as suggested by App. C. At inference time, ROI extraction adds approximately 12 ms per step in our implementation.

8 Conclusion

We presented Seeker, an action-supervised module that learns where visual evidence is needed for visuomotor control. Seeker turns observation–action data into a progression-aware ROI, exposing an explicit spatial bottleneck without relying on semantic boxes, gaze, language grounding, or affordance annotations. Across simulation and real-robot experiments, the same bottleneck improves data efficiency, enables targeted background augmentation, and transfers from RGB crops to point-cloud filtering. These results point to action-grounded visual bottlenecks as a practical interface between perception and policy learning, and more broadly highlight that the visual structure recovered from action supervision depends on the coverage and variation present in the demonstrations.

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Attention from Action, for Action

APPENDIX

This appendix provides implementation details, diagnostics, and extended results for Seeker. App. A details Seeker training and ROI extraction. App. B summarizes the simulation tasks. App. C studies data efficiency through spatial coverage. App. D reports full simulation and point-cloud results. App. E provides real-world setup details and qualitative visualizations.

A Seeker Training Details

A.1 Head Gating Visualizations

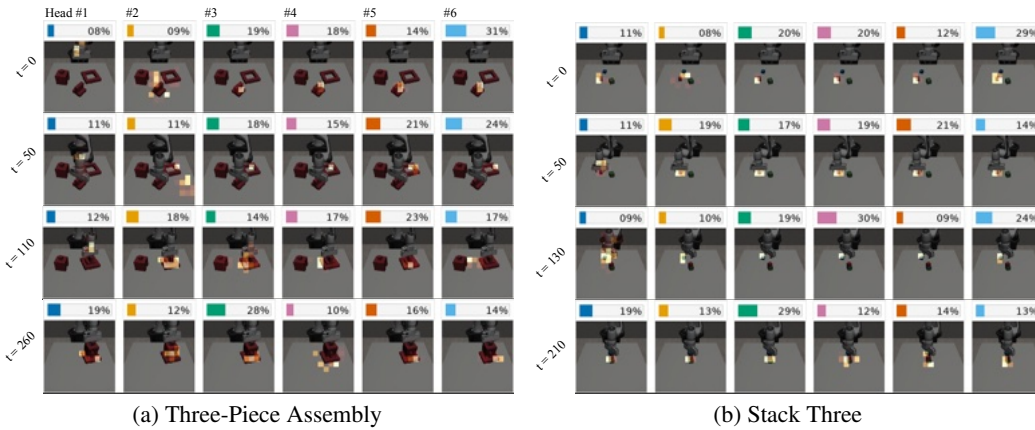


Figure A.1: Per-frame head overlays and corresponding gating weights for two tasks.

As shown in Fig. A.1, the contribution of each attention head changes over the course of the task. Taking the Three-Piece Assembly example, Head 1 remains relatively diffuse and becomes most salient for the final piece. During the final placement stage, Head 4 becomes more diffuse, while Head 3 becomes the dominant contributor.

A.2 Seeker Implementation Details

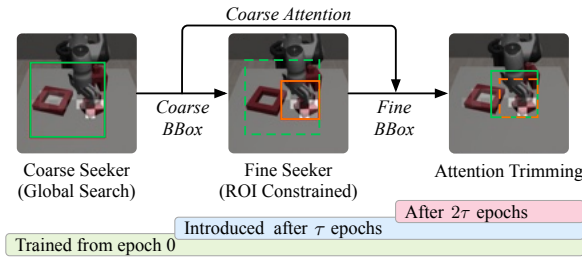


Figure A.2: **Coarse-to-fine Attention Trimming.** From epoch 0, Seeker produces a coarse box; after τ epochs, the fine branch predicts a fine box within the coarse ROI; after 2τ , attention trimming suppresses out-of-box mass and tightens the coarse box.

Coarse-to-Fine Attention Trimming. Fig. A.2 shows that action supervision already steers Seeker attention toward task-relevant regions, but the resulting maps can remain heavy-tailed, yielding non-compact ROIs, especially in long-horizon tasks with shifting targets. We adopt a coarse-to-fine staged training scheme (Fig. A.2) with stage stride τ (epochs) to obtain compact, adaptive ROIs. We train a **coarse** Seeker from the first epoch for localization, introduce a **fine** Seeker after τ epochs that operates only on patch tokens within the coarse ROI (box-cropped token set), and train both branches jointly with shared DINOv3 patch features but separate diffusion heads. After 2τ epochs, we activate

attention trimming: letting $\mathcal{M}_{\text{box}}^{\text{fine}} \in \{0, 1\}^{N_p}$ be the binary patch mask induced by the fine ROI box (1 inside, 0 outside), we form a trimmed target attention for each head by masking and renormalizing the coarse last-step attention $A_h^{(T)}$, and minimize a head-weighted KL divergence,

$$\hat{A}_h = \text{stopgrad} \left[\text{Norm} \left(\mathcal{M}_{\text{box}}^{\text{fine}} \odot A_h^{(T)} \right) \right], \quad \mathcal{L}_{\text{trim}} = \sum_{h=1}^H \omega_h^{(T)} D_{\text{KL}} \left(\hat{A}_h \parallel A_h^{(T)} \right).$$

which suppresses residual out-of-box mass and tightens the coarse ROI until it largely overlaps the fine ROI. After training, we discard the fine branch and retain only the coarse branch for single-pass ROI prediction at inference.

Iterative Refinement. Iterative refinement helps Seeker form task-progress-aware attention. A single state-conditioned query can already localize useful regions because proprioception and gripper state provide strong phase cues, such as picking versus insertion in Square D2. However, similar end-effector states can still correspond to different visual targets depending on object pose, scene layout, or task progress. In such ambiguous cases, a single lookup can produce less compact attention, while refinement lets the query update from retrieved visual evidence before producing the final attention map. We use two query updates in all experiments, and derive the training attention map and pooled context after the second update.

The low-dimensional inputs consist of end-effector translation, end-effector rotation, gripper state, a one-hot embodiment identifier, and a task embedding. Task embeddings are derived from CLIP [30] given short task descriptions; the language serves only as an identifier, does not inject linguistic prior, and is similar to task-specific learnable tokens. Seeker fuses the task embedding with the remaining low-dimensional signals to form its conditioning context (Eq.3.1), while the policy head concatenates the low-dimensional inputs with a down-projected task embedding. In multi-embodiment training, ROI localization degrades without the embodiment identifier, so we include it in both Seeker’s conditioning context and the policy head. For Seeker, we use only the 3D translation and gripper state, since end-effector rotation is largely redundant for coarse ROI localization and can hinder convergence; rotation can be added back for tasks where wrist orientation is essential. Dataset **images** are stored at 240×240 and randomly cropped to 224×224 during training, which is important for stable policy learning [31]. During **mask extraction**, we drop the two lowest-scored attention heads, renormalize the remaining weights to sum to one, and use nucleus sampling ($\text{top_p} = 0.8$) in both simulation and real-robot experiments to suppress low-probability patches. This hard dropping is applied only at extraction time; during policy training, head weights remain soft to preserve differentiability. **Hyperparameters.** Seeker uses the same hyperparameters as the default Diffusion Policy [23]. Learning rate warmup set for diffusion policy is also important for stable ROI emergence, as initial updates without warmup can impose strong early attention biases and hinder convergence.



Figure A.3: **Effectiveness of Pixel Perturbation.** At early training stages (10k updates), Seeker trained without Random Overlay exhibits noisy attention on spurious visual artifacts.

Pixel-level Perturbation. We find pixel-space perturbations crucial for the quality of Seeker’s ROI (Fig. A.3). We therefore apply a simple yet effective Random Overlay augmentation, largely following MirrorAug [17]. At each update, each sample is independently overlaid with probability p_{ov} (i.e., p_{ov} is the per-sample augmentation probability). MirrorAug ramps p_{ov} linearly from 0 to 0.5 during a warmup phase. In contrast, we set the warmup to zero and use $p_{\text{ov}} = 0.5$ from the start of training, so that on average half of the samples in every batch are augmented. Empirically, gradual ramp-up weakens the regularization effect when early spurious biases form, reducing its ability to prevent attention collapse. The same perturbation is also applied when training the RVT-2-style heatmap predictor. We mix the original image I with a random texture image I_{tex} using overlay strength $\alpha = 0.6$:

$$\tilde{I} = \alpha I + (1 - \alpha) I_{\text{tex}} = 0.6 I + 0.4 I_{\text{tex}}.$$

View Complementarity and Sequential Training. In multi-view settings (third-person and eye-in-hand, with proprioception), the most informative view can change over the course of an episode. In particular, the eye-in-hand view often becomes dominant during the interaction phase, while the third-person view can become less informative. Under joint multi-view training, we observe that the third-person ROI may occasionally shift toward the next target or other salient regions. This behavior reflects Seeker exploiting cross-view complementarity, rather than a failure of localization.

However, when a standalone third-person ROI is required (e.g., third-person-only policy or cropping for point clouds), this implicit cross-view dependency is no longer available and can lead to degraded performance. We therefore incorporate additional views sequentially during training. We first train Seeker using only the third-person view and cache the resulting third-person context features. We then initialize a multi-view Seeker that takes the cached third-person features while learning to incorporate the additional view(s). This yields the Seeker variant used in our simulation and real-robot experiments, encourages the third-person representation to remain reliable throughout an episode, and allows adding new cameras without retraining previously incorporated views.

Seeker Localizes Rather Than Controls. Although Seeker’s ROI localization emerges from the policy objective, its pooled context is not intended to serve as a standalone action representation. Directly rolling out actions from Seeker’s context achieves near-zero average success across tasks, with Stack-Three being the highest at 20%, indicating that Seeker captures mainly *where* to attend while discarding fine-grained geometric and contact cues needed for control. Fine-tuning DINO improves this context but remains non-competitive with full policy baselines and weakens the pretrained geometric structure used for stable attention, so we keep Seeker frozen and use it only as an explainable visual bottleneck.

Training Objectives. We compare Seeker trained with different action objectives, including behavioral cloning (BC, direct regression), flow matching [27], and Implicit Maximum Likelihood Estimation (IMLE) [29]. As shown in Fig. A.4, all objectives bias attention toward foreground regions, but produce different ROI structures. BC yields heavier-tailed ROIs with more scattered noise, likely due to pointwise regression under multimodal expert behavior. IMLE produces cleaner maps but often spreads attention over multiple plausible foreground cues. Flow matching yields ROIs closest to the default diffusion-trained Seeker, consistent with its closely related generative formulation.

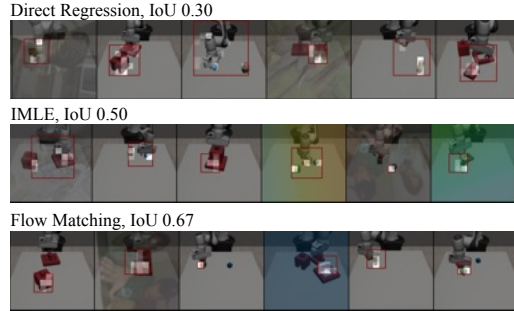


Figure A.4: **Seeker ROI under Different Training Objectives.** Visualizations are shown with the mean IoU to the diffusion-trained Seeker.

We quantify these differences by measuring ROI IoU against the diffusion-trained Seeker on the merged six-task simulation dataset with 100 demonstrations per task. BC, IMLE, and flow matching achieve mean IoUs of 0.30, 0.50, and 0.67, respectively, matching the qualitative trend in Fig. A.4.

B Simulation Task Overviews

We use six simulation tasks from MimicGen [24] and the corresponding released datasets. We re-render the trajectories at higher image resolution (from 84×84 to 240×240) to match DINO’s default input size. Five tasks use a Franka Panda arm and one uses a Sawyer arm (Pick and Place D0). All tasks share a 7D action space, consisting of 3D translation, 3D rotation, and a 1D gripper command. Each task provides a third-person (agentview) camera and an eye-in-hand camera. Task descriptions are summarized below.

Full Workspace Shuffling. Objects are randomized (both rotation and translation) across the workspace. *Stack Three D1*: 6 subtasks (3 pick-and-place). Sequentially stack three cubes. *Square D2*: 2 subtasks (pick-and-insert). Insert a square nut onto a square peg. *Three-Piece Assembly D2*: 6 subtasks (3 pick-and-place). Sequentially assemble three pieces, requiring precise orientation and placement. *Threading D2*: 2 subtasks. Pick a needle by its cubic handle and insert the tip into a hole on a fixture.

Constrained Object Shuffling. Each component is randomized within a constrained region. *Coffee Preparation D1*: 7 subtasks. Pick up the mug, place it on the coffee machine, open the capsule slot’s lid, open the drawer, pick up a capsule, place it into the slot, and close the lid. The drawer has no spatial variation. *Pick and Place D0*: 8 subtasks. Sequentially place four objects into their corresponding baskets. Objects start on the left, and the target baskets are fixed on the right.

C Data Efficiency and Spatial Coverage

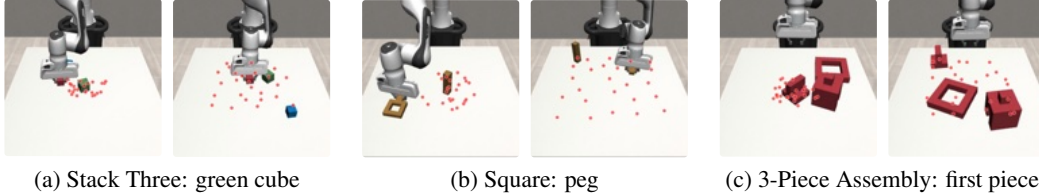


Figure A.5: **Sub-task Spatial Coverage.** Sub-task target distributions used to construct coverage-controlled 25-demo equivalent Seeker pretraining subsets. Each panel shows one representative sub-task, comparing the least diverse subset on the left with the most diverse subset on the right.

This section contextualizes Seeker’s training overhead with a task-specific coverage diagnostic. Our main experiments train Seeker on the full 100-demo policy-training set, giving a conservative estimate of total training cost. Here, we ask whether this cost can be reduced when the selected demonstrations still preserve the spatial variation needed for ROI learning.

We use task-specific Seekers to isolate spatial coverage from the additional factors introduced by multi-task training. In a multi-task Seeker, the effective data requirement is shaped not only by per-task coverage, but also by task balance and inter-task interference, especially when the pretraining set is reduced. Our early experiments showed that multi-task Seeker needed more data to match isolated task-specific Seekers. We therefore treat this study as a controlled diagnostic of spatial coverage, rather than a low-data multi-task pretraining protocol.

We construct coverage-controlled subsets using privileged sub-task annotations only for selection. Each task is decomposed into sub-tasks, and demonstrations within each sub-task are ranked by farthest-point sampling over target poses. An N -demo equivalent setting denotes the top or bottom N demonstrations selected independently per sub-task from the 100-demo pool, rather than N trajectories selected globally. Seeker is trained on the corresponding demonstration windows, with a fixed-range normalizer to avoid subset-specific normalization shifts.

Fig. A.5 visualizes the least and most diverse 25-demo equivalent subsets for representative sub-tasks. Tab. A.I shows that task-specific Seeker is robust to reduced pretraining data when spatial coverage is preserved: Most-25 is within 0.4 points of Full-100 on average, and Most-50 matches Full-100. Least-25 remains competitive but drops by 3.6 points, mainly on Square and 3-Piece Assembly. This suggests that spatial distribution matters most when the pretraining budget is small, especially for tasks such as 3-Piece Assembly where similarly colored, shape-dependent objects induce broader pose-dependent contact regions.

Coverage	Stack Three D1	Square D2	3-Piece Assembly D2	Avg.
Full-100	81.3$\pm$1.2	46.0$\pm$7.2	58.7$\pm$2.3	62.0
Least-25	81.3 $\pm$ 1.2	40.7 $\pm$ 1.2	53.3 $\pm$ 2.3	58.4 _{3.6} ↓
Most-25	78.7 $\pm$ 4.2	44.7 $\pm$ 3.1	61.3 $\pm$ 2.3	61.6 _{0.4} ↓
Most-50	80.0 $\pm$ 2.0	45.3 $\pm$ 7.6	60.7 $\pm$ 5.0	62.0

Table A.I: **Effect of Seeker Pretraining Spatial Coverage.** Success rates (%) are computed as the mean over three seed-level best checkpoints. Full-100 denotes the Seeker used in the main experiments. Least-25 and Most-25 use the least and most spatially diverse 25-demo equivalent subsets, respectively, while Most-50 uses the most spatially diverse 50-demo equivalent subset. Average subscripts report the change relative to Full-100.

Overall, this diagnostic suggests that task-specific Seeker pretraining can be substantively reduced with spatially covered subsets. In the multi-task setting, however, the reducible overhead depends on preserving both spatial coverage and task balance while mitigating cross-task interference. Quantifying this trade-off is beyond the scope of this diagnostic.

D Experimental Details and Results

D.1 RGB Cropping

	Stack Three	Square	3-P Assembly	Coffee Prep.	Pick & Place [†]	Threading
<i>Input-level Comparison</i>						
DiffPo (Pre)	52.7 $\pm$ 5.4	22.0 $\pm$ 3.5	14.0 $\pm$ 9.2	65.3 $\pm$ 2.1	20.0 $\pm$ 5.3	20.7 $\pm$ 2.1
MirrorAug	68.0 $\pm$ 3.3	32.7 $\pm$ 2.5	21.0 $\pm$ 1.0	62.0 $\pm$ 2.0	20.7 $\pm$ 3.1	18.7 $\pm$ 1.2
RVT2-Crop	71.3 $\pm$ 1.2	26.0 $\pm$ 2.0	26.7 $\pm$ 1.2	69.3 $\pm$ 2.3	49.3 $\pm$ 3.1	12.7 $\pm$ 1.2
Seeker (ours)	81.3 $\pm$ 1.2	46.0 $\pm$ 7.2	58.7 $\pm$ 2.3	82.0 $\pm$ 2.0	69.3 $\pm$ 1.2	38.0 $\pm$ 2.0
<i>External SOTA and Privileged ROI References</i>						
RAVEN	80.7 $\pm$ 2.1	50.0 $\pm$ 6.9	26.7 $\pm$ 7.3	72.7 $\pm$ 7.3	54.7 $\pm$ 1.2	28.0 $\pm$ 3.5
Oracle ROI*	80.7 $\pm$ 1.2	44.7 $\pm$ 4.2	62.7 $\pm$ 1.2	81.3 $\pm$ 4.2	71.3 $\pm$ 1.2	44.7 $\pm$ 1.2

Table A.II: **Data Efficiency on MimicGen** [24]. Success rate (%) on six tasks with 100 demonstrations, averaged over three seeds and reported as mean $\pm$ std. RAVEN and DiffPo (Pre) results are taken from the RAVEN paper [19] where protocol-matched. Pick & Place uses our strict episode-success reruns.

D.2 Point-cloud Filtering

Implementation. For each observation, we unproject calibrated depth images from all cameras into a shared world-frame point cloud. Given a Seeker bounding box from camera $\tilde{c}$, we reproject all world-frame points into the image plane of $\tilde{c}$ and retain only points whose projected pixel coordinates fall inside the box. This yields the Seeker-cropped point cloud while still leveraging points from all cameras. Unlike cropping each camera image independently before 3D projection, this strategy uses a single Seeker prediction and does not require training camera-specific Seekers. For the remaining point-cloud experiments, we follow EquiDiff’s [18] default DP3 [25] setup.

<i>Manual</i>	$\times$	$\checkmark$	$\times$	$\checkmark$
<i>Seeker</i>	$\times$	$\times$	$\checkmark$	$\checkmark$
Stack Three D1	0.0 $\pm$ 0.0	23.3 $\pm$ 1.1	34.6 $\pm$ 5.7	47.3$\pm$8.0
Square D2	0.0 $\pm$ 0.0	6.0 $\pm$ 1.1	8.6$\pm$1.1	7.3 $\pm$ 1.1
3-Piece Assem. D2	0.0 $\pm$ 0.0	1.3 $\pm$ 1.1	6.0$\pm$2.0	6.0 $\pm$ 0.0
Coffee Prep. D1	0.0 $\pm$ 0.0	22.0 $\pm$ 6.9	42.0 $\pm$ 6.0	64.3$\pm$8.0
Pick & Place D0	0.0 $\pm$ 0.0	1.3 $\pm$ 1.1	12.6 $\pm$ 1.1	26.0$\pm$3.5
Threading D2	0.0 $\pm$ 0.0	18.0 $\pm$ 2.0	22.6 $\pm$ 4.6	28.6$\pm$4.2

Table A.III: **Full Point-cloud Results.** We compare different point-cloud cropping strategies across all tasks for three seeds, reporting the mean and standard deviation.

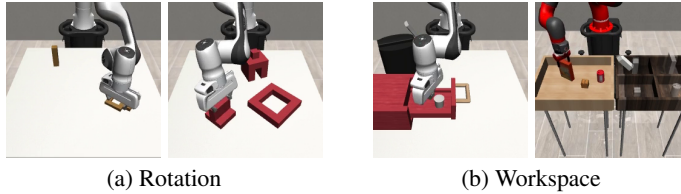


Figure A.6: **Point-cloud Policy Stress Cases.** Under combined cropping, failures mainly arise from (a) inaccurate orientation estimation and (b) sparse task-relevant points in large workspaces, making fine-grained motions difficult to infer.

Failure Analysis. Tab. A.III reports full results across all tasks and seeds. Seeker plus manual workspace cropping performs best in five out of six tasks, but point-cloud policies still lag image-based policies on most tasks. Fig. A.6 shows two typical failure modes. First, DP3 can struggle with fine-grained rotation inference, so Seeker’s spatial focus does not fully resolve orientation ambiguity in the point-cloud representation. This is most pronounced in Three-Piece Assembly, which has broad up-axis rotation variation, and less severe in Coffee Preparation, where orientation variation is smaller. Second, in large workspaces such as Pick & Place, task-relevant points can remain sparse after cropping, especially when Seeker under-crops and leaves distractor points.

D.3 Random Overlay and Guided Overlay

In Sec. 4 we introduced Random Overlay as one of the baseline augmentation techniques, with Seeker’s guided augmentation. We detail them below:

Random Overlay is applied by blending the input image I with a randomly sampled background I_{tex} , $\tilde{I} = (1 - \alpha)I + \alpha I_{\text{tex}}$, where $\alpha \in [0, 1]$ is the constant blending coefficient. Following App. A.2, the augmentation is applied with probability p , linearly warmed up to 0.5 over the first 2500 training steps, with α fixed to 0.6.

Guided Overlay. Guided Overlay follows the same procedure, but replaces the global blending coefficient with a spatially varying map derived from Seeker’s mask M . The mask is produced at patch resolution (e.g., 14×14 for a 224×224 input), and we bilinearly upsample it to the image resolution before blending. The image I is either the ROI-cropped third-person view or the full eye-in-hand view. We convert M (summing to one) into an image-valued map by $M_{\text{norm}} = M / \max(M)$ and clamp it to $[0.3, 0.8]$ to obtain M_{clamp} . We then blend per pixel as

$$\tilde{I} = M_{\text{clamp}} \odot I + (1 - M_{\text{clamp}}) \odot I_{\text{tex}},$$

Clamping is necessary as Seeker’s mask is produced at patch resolution rather than pixel-level accuracy. Overly extreme blending that fully replaces low-confidence regions or preserves high-confidence regions can degrade performance. Clamping moderates these extremes: low-confidence regions are strongly perturbed but still partially preserved, while high-confidence regions are only mildly perturbed rather than fully protected.

E Real Experiments

E.1 Tasks and Setup

All tasks are performed on a UFactory xArm 7 with fixed third-person and eye-in-hand RGB cameras. Demonstrations are collected via teleoperation, and objects are randomly shuffled within the workspace during both data collection and policy evaluation. All tasks use position-based end-effector control at 10 Hz. Both proprioceptive states and actions are 10-dimensional, comprising a 3D Cartesian end-effector position, a 6D representation of end-effector rotation, and a 1D continuous gripper state. Image observations are captured at 848×480 , center-cropped for the third-person view and square-padded for the eye-in-hand view, then resized to 240×240 . Task descriptions are provided below, with dataset statistics summarized in Tab. A.IV.

Coffee Transport requires the robot to scoop coffee beans from a yellow bowl using a spoon and pour them into an espresso cup. This task requires precise control of tool orientation and pouring motion, with the small white espresso cup placed against a white-dominant textured background, making target localization brittle from RGB observations. Successful execution requires coordinated reaching, scooping, transporting, and controlled tilting without spilling.

Table Cleanup requires the robot to sequentially grasp and insert two toy vegetables with different geometries, a long thin carrot and a round tomato, into a bin, followed by placing a lid on top. The task involves multiple object interactions, ordered subtasks, and non-trivial 3D rotations during insertion, requiring temporal consistency and accurate pose control across subtasks.

Board Assembly requires the robot to sequentially pick three pieces with distinct base shapes, round, square, then hexagonal, and insert them into corresponding slots on a board. Although the geometries match exactly, the part colors intentionally do not correspond to the slot colors, increasing the reasoning difficulty. Successful execution requires millimeter-level accuracy and robust shape-based pose estimation across a long horizon.

Task	# Demos	# Subtasks	ImgD	PropD	ActD
<i>Coffee Transport</i>	50	4	$2 \times 3 \times 240 \times 240$	10	10
<i>Table Cleanup</i>	100	6	$2 \times 3 \times 240 \times 240$	10	10
<i>Board Assembly</i>	100	6	$2 \times 3 \times 240 \times 240$	10	10

Table A.IV: **Real-world task summary.** # Demos: demonstrations used for both Seeker and downstream policies; ImgD: input RGB observation dimensionality; PropD and ActD: proprioception and action dimensions.

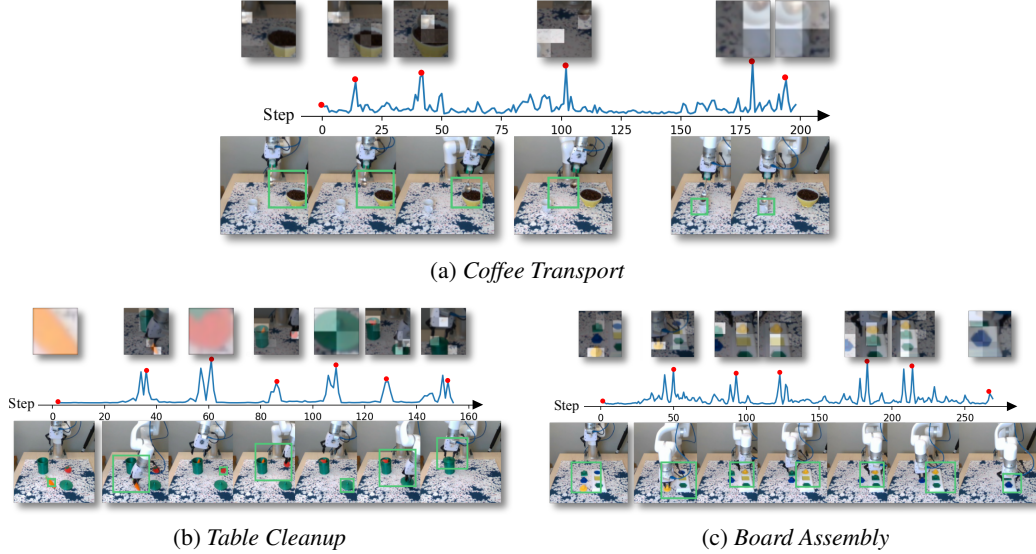


Figure A.7: **Seeker-selected Keyframes for Real-world Tasks.** Predicted ROIs, crops, and masks are aligned with peaks in pairwise cosine distance from Seeker’s context feature.

E.2 Training Details

We train our methods and all baselines on real-world data using the same protocol as in simulation. Limited demonstrations and complex visual backgrounds make convergence more challenging, especially with image-space noise from Random Overlay for baselines and Guided Overlay for Seeker policies. We therefore adjust two hyperparameters: the overlay warm-up is increased to 5,000 steps for all policies, and Seeker training is extended to 600 epochs.

E.3 Seeker Visualization

Fig. A.7 shows additional real-world visualizations, including Seeker-selected keyframes, masks, and crop outputs. Keyframe scores are computed from the cosine distance between consecutive Seeker context vectors, as in simulation, and highlight meaningful stage transitions. Qualitatively, Seeker maintains task-relevant localization and progression-aware focus switching on real hardware.

Compared with MimicGen [24], real-world demonstrations are generated by human operators and contain smoother continuous translation and rotation within each subtask. Accordingly, Seeker more often exhibits an *interaction-centric* focus, covering both the manipulated object and the target, such as the spoon tip and cup in *Coffee Transport* or the part and slot in *Board Assembly*. In *Table Cleanup*, where manipulation involves less rotation, Seeker more often follows an *object-centric* pattern. Overall, these results suggest that Seeker can switch between *object-centric* and *interaction-centric* bottlenecks depending on the task and data distribution.